

# Chapter 9

## A Late Nuclear Proto-Indo-European verbal type: Intransitive thematic nasal-infix present actives

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This paper examines primary verbal formations in the ancient Indo-European languages that also play a role in deadjectival derivation. It builds on recent work on the historical morphology of the intransitive change of state nasal-affix thematic active presents in the “Northern Indo-European” languages – Baltic, Slavic, and Germanic – which continue an anticausative to the well-known athematic nasal-infix transitive presents. The paper argues that this formation has exact cognates in Ancient Greek, Vedic Sanskrit, and Indo-Iranian, more generally, where we find morphologically identical presents that as in the Northern Indo-European languages make semantically intransitive change of state formations to both primary verbs and root-derived property concept adjectives. The agreement across these branches supports the reconstruction of this formation for late Nuclear Indo-European, as a continuation of a deeper Proto-Indo-European type.

### 1 Introduction

One of the more remarkable features of Indo-European (IE) morphology is that deadjectival verbal formations are partially handled with dedicated denominative morphology and partially with the root-derived – “primary” – morphology that otherwise holds across Proto-Indo-European (PIE) verbal classes. While the deadjectival denominative formations have been thoroughly studied and are well



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understood, the insight that primary formations also played a role in deadjectival derivation – at least in the sphere of property concept adjectives – is a recent discovery whose full parameters have yet to be elucidated.<sup>1</sup>

This paper aims to further advance this discussion and takes its start from recent work on the historical morphology of the intransitive change of state nasal-affix thematic active presents in the “Northern Indo-European” languages (Baltic, Slavic, and Germanic), whose origins have been suggested to continue a morphologically peculiar anticausative to the well-known athematic nasal-infix transitive presents. The paper argues that this type has exact cognates in Ancient Greek and Vedic Sanskrit (and Indo-Iranian, more generally), where we find morphologically identical presents that as in the Northern Indo-European languages make semantically intransitive change of state formations to both primary verbs and subsets of root-derived property concept adjectives. Based on the agreement of these branches, it is possible to reconstruct the formation for late Nuclear Indo-European<sup>2</sup> and to project its origins deep into PIE.

## 2 Northern Indo-European

The starting point for this investigation is the magisterial 2007 Harvard PhD thesis of Yaroslav Gorbachov (Gorbachov 2007; see further Villanueva Svensson 2006, 2011) on the inchoative nasal-affix presents in Baltic, Slavic, and Germanic, the Northern Indo-European languages. This nasal-affix verbal type is well attested in all three Northern Indo-European branches, where it generally serves to make 1) intransitive change of state/location formations – often categorized as “inchoatives” – to intransitive statives, less frequently activity verbs, 2) anticausatives to causative formations, and 3) denominatives to property concept adjectives and, more rarely, substantives. For a survey of forms across all representative semantic categories, cf. the examples in (1).

- (1) Inchoative nasal-affix presents in Slavic, Baltic, and Germanic
  - a. Old Church Slavonic (OCS)
    - *sędŏ, sesti, sędŭ* ‘sit down’
    - *vŭzlēgŏ, -lešti, -legŭ* ‘lie down’

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<sup>1</sup>See Rau 2009 with lit., 2013, 2016, 2017, and further Nussbaum 2022.

<sup>2</sup>“Nuclear Indo-European” is used to refer to the subgroup of Indo-European languages that shared innovations after the departure of Anatolian from the PIE speech community. I use “late Nuclear PIE” to refer to the subgroup of Nuclear PIE languages that remain after the departure of Tocharian and then later Proto-Italo-Celtic.

- *stanŋ, stati, staxŭ* ‘stand up’
- *pri/vŭznikŋ, -nikŋti, -nikŭ* ‘spring up’
- *promŭknŋ, -mŭknŋti sę, -mŭkŭ* ‘go out, spread’
- *prilŭ(p)ŋ, -lŭ(p)ŋti, -lŭpŭ* ‘cleave, cling to’
- *uglibŋ, -glibŋti, -glibŭ* ‘get stuck’
- *izbęgnŋ, -bęgnŋti, -bęgŭ* ‘flee, avoid’
- *postignŋ, -stignŋti, -stigŭ* ‘come upon, reach’
- *vŭzbŭ(d)ŋ, -bŭ(d)ŋti, -bŭdŭ* ‘wake up’
- *usŭ(p)ŋ, -sŭ(p)ŋti, -sŭpŭ* ‘fall asleep’
- *navykŋ, navyknŋti, navykŭ* ‘get used to’
- *oslŭpnŋ, -slŭpnŋti, -slŭpŭ* ‘go blind’ (: *slępŭ*)
- *ugasŋ, -gasŋti, -gasŭ* ‘go out, be extinguished’
- *prisvędnŋ, -svędnŋti, -svędŭ* ‘wilt’
- *omrŭknŋ, -mrŭknŋti, -mrŭkŭ* ‘grow dark’ (: *mŭrkŭ\**)
- ORuss. *svŭ(t)nuti* ‘grow light’
- *sŭ/pomrŭznŋ, mrŭznŋti, -mrŭzŭ* ‘freeze, congeal’ (: *mrazŭ*)
- *sŭxnŋ, sŭxnŋti, sŭxŭ* ‘dry up’ (: *suxŭ*)

**b. Baltic**

- OPr. *sŭnda-* ‘sit down’
- OPr. *polŭnka, polŭikt* ‘stays’ (: Lith. dial. *-liŭka* ‘remains’)
- Lith. *taŭpa, tŭpti, tŭpo* ‘become’
- Lith. *kniŭba, knŭbti, knŭbo* ‘sink, fall’
- Lith. *kaŭka, kŭkti, kŭko* ‘start out, go’
- Lith. *muŭka, mŭkti, mŭko* ‘break free, escape’
- Lith. *liŭpa, lipti, lipo* ‘cling, stick to, climb’
- Lith. *pliŭta, plisti, plito* ‘spread out’
- Lith. *skiŭda, skisti, skido* ‘fall apart, split asunder’
- Lith. *atbuŭda, atbŭsti, atbŭdo* ‘wake up’
- Lith. *miŭga, mŭgti, mŭgo* ‘fall asleep’
- Lith. *supraŭta, suprŭsti, suprŭto* ‘understand’
- Lith. *patiŭka, patŭkti, patiko* ‘suits, fits’
- Lith. *šviŭta, švisti, švito* ‘dawn’
- Lith. *sniŭga, snŭgti, snŭgo* ‘snow’
- Lith. *tuŭka, tŭkti, tŭko* ‘grow fat’ (: *taukai*)
- Lith. *pŭva, pŭti, pŭvo* ‘rot, decay’
- Lith. *sŭsa, sŭsti, sŭso* ‘dry up’ (: *saŭsas*)
- Lith. *dumba, dŭbti, dŭbo* ‘grow hollow’ (: *dubŭs*)

c. Germanic

- Go. *standan* ‘stand’, ON *standa* ‘id.’
- Go. *aflifnan* ‘remain’, ON *lifna* ‘id.’
- OE *climban* ‘climb’, OHG *klimban* ‘id.’
- Go. *rinnan* ‘run’, ON *renna* ‘id.’
- Go. *aflinnan* ‘leave off, desert’, ON *linna* ‘cease, stop’
- Go. *wiknan* ‘start to give in, yield’, ON *vikna* ‘id.’
- Go. *usgutnan* ‘spill out’
- Go. *usbrikan* ‘break off’
- Go. *gawaknan* ‘wake up’, ON *vakna* ‘id.’
- ON *sofna* ‘fall asleep’
- Go. *ga(h)nipnan* ‘grow sad’, ON *hnipna* ‘become downcast, droop’
- Go. *gawairþnan* ‘become reconciled’
- Go. *gathaursnan* ‘get dry’, ON *þorna* ‘id.’
- Go. *gastaurknan* ‘grow stiff, rigid’, ON *storkna* ‘id.’ (: ON *starkr*)
- Go. *bulgnan* ‘swell up’, ON *bolgna* ‘id.’
- ON *fúna* ‘rot’
- Go. *gafullnan* ‘grow full’, ON *fullna* ‘id.’ (: *fulls*)
- ON *hvítna* ‘goes pale’ (: *hvítr*)
- ON *róðna* ‘turn red’ (: *rauðr*)

This Northern Indo-European verbal type also shows a large number of two- and three-way cognate sets among the languages – cf. the forms in Table 1.

In his thesis, Gorbachov makes three important observations about this type and its prehistory. First, he notes that these verbs inflect as *active* and *thematic* presents with a nasal affix across all three branches – so in Gothic and residually elsewhere in Germanic –, and that this active thematic inflectional pattern should be reconstructed for the ancestor of the type in all three languages. See Gorbachov 2007 *passim*, with discussion of alternative analyses. Second, he points out that the type systematically corresponds to transitive, athematically inflected nasal-infix presents in other IE languages, and is ultimately to be aligned with this transitive type – see, e.g., the forms in Table 2.

Third, he argues that based on the function of the formation, its active thematic inflection, and its correlation with transitive athematic nasal-infix verbs in the other IE languages, it is possible to set up a descriptive derivational process for the ancestor of these branches, whereby an anticausative is made to the transitive athematic nasal-infix present by putting the stem in the zero-grade, suffixing the

Table 1: Cognate inchoative nasal-affix presents in Slavic, Baltic, and Germanic

| OCS                                                     | Baltic                                                    | Germanic                                               |
|---------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------|
| <i>sędę, sesti, sędŭ</i> ‘sit down’                     | OPr. <i>sīnda-</i> ‘sit down’                             |                                                        |
| <i>prilī(p)nę, -lī(p)nęti, -līpu</i> ‘cleave, cling to’ | Lith. <i>līm̃pa, lipti, lipo</i> ‘cling, stick to, climb’ | Go. <i>aflifnan</i> ‘remain’                           |
| <i>promŭknę, -mŭknęti sę, -mŭkŭ</i> ‘go out, spread’    | Lith. <i>muñka, mŭkti, mŭko</i> ‘break free’              |                                                        |
| <i>uglībņę, -glībņęti, -glībŭ</i> ‘get stuck’           |                                                           | OE <i>climban</i> ‘climb’,<br>OHG <i>klimban</i> ‘id.’ |
| <i>vŭzbŭ(d)nę, -bŭ(d)nęti, -bŭdŭ</i> ‘wake up’          | Lith. <i>atbuñda, atbŭsti, atbŭdo</i> ‘id.’               |                                                        |
| <i>usŭ(p)nę, -sŭ(p)nęti, -sŭpŭ</i> ‘fall asleep’        |                                                           | ON <i>sofna</i> ‘fall asleep’                          |
| ORuss. <i>svī(t)nuti</i> ‘grow light’                   | Lith. <i>šviñta, švisti, švito</i> ‘dawn’                 | ON <i>hvitna</i> ‘goes pale’                           |

accented thematic vowel, and inflecting the formation as an active, as outlined in Table 3.

In terms of its deeper PIE origins, Gorbachov (2007: 200–214) argues that the formation is to be traced to a  $*h_2e$ -conjugated version of the athematic nasal-infix present, which ultimately served to make anticausatives and which as all  $*h_2e$ -conjugation active present types underwent analogical thematization on the way to NPIE.<sup>3</sup> In my view, Gorbachov is correct in his analysis and reconstruction of this type for late PIE. The intention of this paper is to further support his analysis by outlining additional reflexes of the type in Ancient Greek and Indo-Iranian.<sup>4</sup>

<sup>3</sup>See Jasanoff 2003 for the PIE  $*h_2e$ -conjugation and its systematic thematization in the Nuclear IE languages.

<sup>4</sup>A further important feature of these presents is that in Slavic – the only Northern Indo-European branch to preserve the inherited IE aorist – they systematically correlate with the thematic aorist.

Table 2: Correspondences between inchoative nasal-affix presents in Slavic, Baltic, and Germanic and transitive athematic nasal-infix presents in other IE languages

| OCS                                                                     | Baltic                                                                                                                                                           | Germanic                                                                                       | Nasal-infix pres.                                                                                                                                                |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>prilǐ(p)nq̃</i> ,<br>-lǐ(p)nq̃ti, -lǐpu<br>‘cleave, cling to’        | Lith. <i>liṁpa</i> , <i>lipti</i> ,<br><i>lipo</i> ‘cling, stick<br>to, climb’                                                                                   | Go. <i>aflifnan</i><br>‘remain’                                                                | Ved. <i>limpáti</i> AV+, <sup>2</sup> Gk.<br>λιπαίνω ‘anoint’ Att.<br>Ion.                                                                                       |
| <i>promŭknq̃</i> ,<br>-mŭknq̃ti sę̃,<br><i>mŭkŭ</i> ‘go out,<br>spread’ | Lith. <i>muñka</i> ,<br><i>mŭkti</i> , <i>mŭko</i><br>‘break free,<br>escape’                                                                                    |                                                                                                | Ved. <i>muñcáti</i> ‘release,<br>let go’, Lat. <i>ēmungō</i> , -ere<br>‘blow the nose’                                                                           |
| <i>vŭzbŭ(d)nq̃</i> ,<br>-bŭ(d)nq̃ti, -bŭdŭ<br>‘wake up’                 | Lith. <i>atbuñda</i> ,<br><i>atbŭsti</i> , <i>atbŭdo</i><br>‘id.’<br><br>Lith. <i>skiñda</i> ,<br><i>skisti</i> , <i>skido</i> ‘fall<br>apart, split<br>asunder’ |                                                                                                | OIr. <i>ad-boind</i> ‘give<br>notice’ (Gk. πυνθάνομαι<br>Hom.+)<br><br>Ved. <i>chináti</i> ‘split off,<br>break off’, Lat. <i>scindō</i><br>‘id.’                |
| ORuss. <i>rínut’</i><br>‘stream, flow’                                  |                                                                                                                                                                  | OHG <i>scrintan</i><br>‘split open, burst’<br><br>Go. <i>rinnan</i> , ON<br><i>renna</i> ‘run’ | Ved. <i>kṛntáti</i> ‘cut, split’,<br>YAv. <i>kəṛənta-ti</i> ‘id.’<br><br>Ved. <i>riṇáti</i> ‘set in<br>motion, swirl’, Gk.<br>ὀρίνω ‘stir, move excite’<br>Hom.+ |

Table 3: Derivational relationship between athematic nasal-infix transitives and thematic nasal-infix intransitives

| Athematic Nasal-Infix Tr.                                                                                                       |   | Thematic Nasal-Infix Intr.                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| * <i>munék-ti</i> / <i>munk-tói</i> ‘release, set free’<br>Ved. <i>muñcáti</i> , Lat. <i>ēmungō</i> , -ere                      | → | * <i>munk-é-ti</i> ‘go free, escape’<br>OCS <i>promŭknq̃</i> , -mŭknq̃ti sę̃, -mŭkŭ<br>‘go out, spread’, Lith. <i>muñka</i> , <i>mŭkti</i> ,<br><i>mŭko</i> ‘break free’                                |
| * <i>linép-ti</i> / <i>linp-tói</i> ‘make adhere to’<br>Ved. <i>limpáti</i> AV+, <sup>2</sup> Gk. λιπαίνω ‘anoint’<br>Att. Ion. | → | * <i>linp-é-ti</i> ‘cling, stick to’<br>OCS <i>prilǐ(p)nq̃</i> , -lǐ(p)nq̃ti, -lǐpu ‘id.’,<br>Lith. <i>liṁpa</i> , <i>lipti</i> , <i>lipo</i> ‘cling, stick to,<br>climb’, Go. <i>aflifnan</i> ‘remain’ |

### 3 Ancient Greek

In Ancient Greek (and further Classical Armenian<sup>5</sup>), it is possible to identify several sets of thematic nasal-affix presents that can be aligned with the Northern Indo-European type outlined by Gorbachov – cf. the five types listed in examples (2)–(6).<sup>6</sup>

- (2) *Nu*-presents,<sup>7</sup> e.g., φθίνω/φθίνω ‘decay, waste away, disappear’ Hom.+; φθινύθω Hom.+ (tr., intr.), ἔφθιτο Hom.+; ἔφθιον Hom.+; ἔφθ(ε)ισα Hom.+; ἔφθιται Hom.+ (: Ved. *kṣiṇāti*, *kṣiṇóti* ‘destroy’ AV, YAv. *jinā-ti* ‘id.’; ON *dvena* ‘disappear, decrease’, ON *dvína* ‘id.’)<sup>8</sup>
- (3) Nasal-infix to sonorant-plus-laryngeal-final roots
- a. θάλλω ‘blossom, flourish’ Hes. *hHom.*+; ἔθαλον *hHom.*+, τέθηλα Hom.+; θαλέθω Hom.+ (: θαλερός Hom.+; ἡ θάλεια Hom.+; τὸ θάλος Hom.+) <sup>C</sup> (: Alb. 1sg. *dal*, *del* ‘go out’ < \**dalnō*, *dalnet* and MW *deillyaw*\* << \**daln-*, see Schumacher 2004: 257–259 and Schumacher & Matzinger 2013: 968–969)
- b. δύνω ‘go into, enter; go down, sink’ Hom.+<sup>A</sup>; δύω, -ομαι Hom.+; ἔδυν Hom.+ (intr.), ἔδυσσα, -αμην Hom.+

<sup>5</sup>The Greek type has a close match in Classical Armenian, where, e.g., presents in *-ane/im* regularly correlate with nasal-infix presents elsewhere, are made to thematic preterites, both inherited thematic aorists and recategorized thematic imperfections, and show up in both transitive and intransitive formations. For the facts and discussion, see Kocharov 2019: 162ff. with lit. The details of this type, whose core likely depends on a prehistoric Greek and Armenian genetic or convergence-related innovation, I leave for discussion elsewhere. For a useful orientational assessment of the relationship between Greek and Armenian, see most recently Kim 2018 with lit.

<sup>6</sup>Superscript <sup>A</sup> marks verbs that at least optionally take an acc. object. Verbs that show strong Caland system associations, which are frequent across these categories, are marked with superscript <sup>C</sup>. Note that all Caland-system-associative verbs can in principle be understood as deadjectival provided they continue to stand in a synchronically transparent relationship with property concept adjectives derived from the same roots. For primary verbs as a foundational element of the PIE and IE Caland system, see Rau 2009, 2013, 2016, 2017, and further Nussbaum 2022.

<sup>7</sup>The poetic verb ἄνω, -ομαι ‘accomplish, finish’ Hom.+ (: ἀνύω ‘id.’ Hom.+) may also belong in this class if it is a relatively late innovation of the sort noted below. Another possibility is that the form is a recategorized thematic aorist, either an inherited formation – viz. *ásanat* RV+ (: *asāniṣam* RV): *sanóti* RV, OAv. subj. *hanāt* – or an inner-Greek remodeling of a root aorist, assuming the root is \**senh*<sub>3</sub>-. This latter analysis may have support in the peculiar aorist ἦνεσα IG.7.3226, Orchomenos and elsewhere, if it represents the contamination product of this thematic aorist and the *s*-aorist otherwise well attested for this verb.

<sup>8</sup>As comparanda, I provide forms that are directly cognate and those that are closely parallel.

- c. θύνω ‘rush, dart along’ (: analogical θύνέω Hes. Sc.; θύω/θυίω ‘rage, seethe’ Hom., ἔθυσσα Call.) (: Ved. *dhūnóti* ‘shake’ to *\*d<sup>h</sup>euH-* [plus θύω/θυίω ‘rage, seethe’ Hom., so LIV<sup>2</sup>], but LIV<sup>2</sup> s.r. to *\*d<sup>h</sup>eu-* because of θύνέω Hes. Sc.[!])<sup>9</sup>
  - d. κάμνω ‘(tr.) produce/acquire through labor/toil (Hom.[+], but rare); (intr.) labor, toil, grow weary toiling’ Hom.+<sup>A</sup>: ἔκαμον Hom., κέκηκα Hom.+ (: ἀκάμας, -αντος Hom.+; *śamniše* ‘labor’ YV, *śamāyāte* ‘id.’ RV, *aśamīt* ‘ist ruhig geworden’ AVP, *ásamanta* ‘labored’ YV)
  - e. ὀφείλω Att. Ion., ὀφέλλω Hom. Lesb. Arc., ὀφίλω Cret. Arc. (Myc.) ‘owe, be obliged to pay for/render; be bound, obliged to (+ inf.)’<sup>A</sup>: ὄφλον Hom. (Myc.), ὄφληκα Hom.+ Dial. (: ὀφλισκάνω Att., etc.; τὸ ὀφελος Hom.+ [Myc.], ὀφλητής [gloss]; cf. LIV<sup>2</sup> Addenda, *\*h<sub>3</sub>b<sup>h</sup>elh<sub>1</sub>-*)
- (4) Nasal-infix to obstruent-plus-laryngeal-final roots
- a. aor. (≡ impf.) ἔχων ‘yawn, gape’ Hom.+; κέχηνα Hom., χάσκω Att. Ion., χαίνω late<sup>C</sup> (: OCS *zinǫti* ‘gape’, OHG *ginēn* ‘yawn’, ON *gína* ‘id.’; LIV<sup>2</sup> Addenda, *\*ǵ<sup>h</sup>eh<sub>2</sub>-* and *i*-present *\*ǵ<sup>h</sup>eh<sub>2</sub>i-*)<sup>10</sup>
  - b. φθάνω/φθάνω ‘(tr.) be beforehand, anticipate; (intr.) come/act first (±part.)’ Hom.+<sup>A</sup>: ἔφθην Hom. Att., φθάμενος Hom. Hes., ἔφθασα Att. Ion. (: Ved. opt. *daghnuyāt* ‘würde knapp verfehlen’ KS, *má dhak* ‘soll nicht verfehlen’ RV. See LIV<sup>2</sup> *\*d<sup>h</sup>eg<sup>uh</sup>h<sub>2</sub>-* for further possible comparanda.)
- (5) Nasal-infix to stop-final root
- a. λάμπω ‘shine’ Hom.+ (: λαμπρός Hom.+; Hitt. *lap<sup>-zi</sup>*, pret. *lāp<sup>-ta</sup>* [cf. Lith. *lópé*, etc.])<sup>C</sup>
  - b. [ὁ θρόμβος ‘lump, clot’ Att. Ion.: τρέφω, -ομαι ‘make firm, congeal’, ἔτραφον Hom. (intr.), ἔτρεψα, τέτροφα Hom.+<sup>C</sup> (: ταρφύς Hom., τάρφα Hom., τὸ τάρφος Hom.+; Lith. *driṁba*, *dribti* ‘fall in flakes, drop’)]<sup>11</sup>

<sup>9</sup>Given the structure of the root, the form is best assigned, *pace* LIV<sup>2</sup>, who align it with the *nu*-class above, to the category of nasal-infix formations to laryngeal final roots, with Ved. *dhūnóti* an analogical innovation.

<sup>10</sup>Here I assume, with LIV<sup>2</sup>, that aor. ἔχων is an aspectually recategorized imperfect. A parallel shift is responsible for perfective aspect of OCS *zinǫti*.

<sup>11</sup>This and the following two nouns obviously depend on an underlying nasal-infix present, either an athematic transitive or a thematic intransitive of the sort under discussion. I assign them to the latter type, first because of the semantic profile of the roots and their derivational *averbo*, which – at least in the case of ὁ θρόμβος and τὸ θάμβος – is exactly parallel to the

- c. [τὸ θάμβος ‘astonishment’ Hom.+; θαμβέω ‘be astonished’ (: ἐθάμβησε Hom.)<sup>A</sup>, ἔταφον Hom., τέθηπα Hom. (: θάψ, θωπός Att. Ion.<sup>C</sup>; deverbal<sup>7</sup> PGmc. *\*dumba-*). Non-nasal root, *pace* Hackstein 2002: 237 with lit.<sup>12</sup>]
- d. [ὁ στρόμβος ‘top; spinning object’ Hom.+; στρέφω, -ομαι ‘twist, turn’ Hom., ἔστρεψα, -αμην Hom.] Cf. YAv. nom. sg. *θrafts* ‘satisfaction (vel sim.)’ Yt.5.26, YAv. *θraftadra-* ‘rich in; satisfied’: Ved. *tṛmpāti* ‘take delight in, enjoy (+gen.)’ RV. See Section 4.
- (6) -άνω to obstruent final roots<sup>13</sup>
- a. ἀνδάνω ‘be pleasing’ Hom.+ (: ἡδομαι Att. Ion. Dial.): ἔαδον Hom., ἔαδα Hom. (: ἡδύς Hom., τὸ ἡδος Hom., etc.<sup>C</sup>; Ved. *s(u)vádati* ‘make sweet, sweeten’ RV, *svádāte* ‘become sweet’ RV)
- b. ἐκφλυδάνει ‘break out (of sores)’ Hp. (: φλυδαρός ‘soft, flabby’ Hp. *ap.* Gal.19.152, φλυδάω ‘have excess of moisture, be soft/flabby’ Hp.)<sup>C</sup>
- c. ἀργάνω ‘be white’ E.+ (: ἀργός Hom., τὸ ἄργος Hom., etc.)<sup>S</sup><sup>C</sup>
- d. οἰδάνω ‘swell (intr.)’ I646 (οἰδάνεται), Ar.Pax.116 (οἰδάνων); ‘make swell’ I554 (= A.R.1.478): οἰδαίνω ‘swell (intr.)’ A.R.+ (tr., late and rare), οἰδέω ‘id.’ Hom. (: οἰδαλέος Hom., τὸ οἶδος Hom., etc.<sup>C</sup>; ClArm. *aytnowm* ‘swell [intr. *tantum*]’)<sup>14</sup>

other verbs that can be assigned to this class, and second, because the retention of the nasal-infix formant in a nominal derivative is more easily explained as the result of a synchronically isolated thematic nasal-infix intransitive – replaced prehistorically in Greek by the -άνω type below – than an otherwise reasonably well-paralleled athematic nasal-infix transitive.

<sup>12</sup>The long vowel in the perfect τέθηπα Hom.+ and the root noun θάψ, θωπός Att. Ion., whose secondary origins are difficult to motivate, make the reconstruction of an underlying non-nasal root the default assumption.

<sup>13</sup>This is a productive present-stem formant in Greek, which is regularly used to generate presents from thematic aorist forms. I mostly omit obviously – e.g., ἀλφάνω Att. (: ἄλφω Hsch.): ἡλφον Hom.+ – and likely – e.g., λυμπάνω Hom. (v.l.) Sa. Th. Hp. (+) (: λείπω Hom.): ἔλιπον Hom.+ – secondary creations. I include some obviously or likely secondary, inner-Greek formations here to illustrate more fully the semantic subcategories and coherence of this subtype. I use <sup>S</sup> to mark such formations as secondary, Greek-internal innovations. Forms are arranged according to semantic subtype, from property concept to perception/cognition and possession, *inter alia*.

<sup>14</sup>Intransitive οἰδάνων at Ar.Pax.116 is an obvious *forma difficilior* and points to an original intransitive meaning for the active. The Homeric opposition transitive active: intransitive middle has been analogically created according to a well-attested Greek morphological restructuring process – cf., e.g., αἶθω ‘burn (intr.)’ Pi.S., αἶθουσα ‘portico’ Hom.+; αἶθω ‘burn (tr.)’, αἶθομαι ‘burn (intr.)’ Hom.+. The intransitive active is also supported by largely intransitive and synchronically/semantically isolated οἰδαίνω ‘swell (intr.)’ A.R.+ (tr., late and rare), which is likely

- e. κῦδάνω ‘be triumphant (intr., Y.42); glorify (tr., ξ73)’: κῦδαίνω ‘glorify, honor (tr. *tantum*)’ Hom.+, ἐκῦδῆνα Hom.+ (: κῦδρός Hom.+, κῦδάλιμος Hom.+, τὸ κῦδος Hom.+, etc.)<sup>C</sup>
- f. λανθάνω, -ομαι ‘escape notice (±acc.); forget’ (: λήθω, -ομαι Hom.+) <sup>A</sup>: ἔλαθον, -ομην Hom.+: ἐλέθηα Hom.+ (: λάθρηι Hom.+, etc.)<sup>\$C</sup>
- g. ὀλισθάνω ‘slip’ Att. (: ὀλισθαίνω ‘id.’ Arist.+): ὥλισθε Hom.+ (: Go. *slindan* ‘schlingen’; Ved. *srédhati* ‘gleitet aus, handelt falsch’ RV, *sridhat* RV)<sup>C</sup>
- h. ἀμαρτάνω ‘miss the mark, go astray’ Hom.+: ἥμαρτον Hom.+
- i. τυγχάνω ‘chance on, happen, hit the mark’ Hom.+: ἔτυχον Hom.+, τετύχηκα Hom.+ (: τεύχω ‘make ready, produce’ Hom.+, ἔτευξα Hom.+, τετευχώς Hom. [Myc.]])
- j. πυνθάνομαι ‘learn, ascertain’ Hom.+ <sup>A</sup> (: πεύθομαι Hom.+): ἐπυθόμην Hom.+: πέπυσμαι Hom.+ (: OCS *vŭzbŭnŭti*, *vŭzbŭde* ‘wake up’, Lith. *atbuñda*, *atbùsti*, *atbùdo* ‘id.’; OIr. *ad·boind*)<sup>C</sup>
- k. αἰσθάνομαι ‘perceive, take notice of’ Att. Ion. <sup>(A)</sup>: ἡίσθόμην Att. Ion., ἡίσθημαι Att. Ion.<sup>\$</sup>
- l. μανθάνω ‘learn, notice, understand’ Pi+ <sup>A</sup>: ἔμαθον Hom.+, μεμάθηκα Att. Ion.<sup>\$</sup>
- m. δαρθάνω ‘fall asleep’ Hierocl.: ἔδαρθον Hom.+<sup>\$</sup>
- n. θιγγάνω ‘touch, handle’ Att. Ion.+: ἔθιγον Archil.+<sup>15</sup>
- o. χανδάνω ‘take hold; be capable’ Hom.+ <sup>A</sup>: ἔχαδον Hom.+, κέχα/ονδα Hom. (: Lat. *prehendō*, -ere ‘seize, grasp’, OIr. *ro·geinn* ‘occupies’, Go. *duginnan* ‘begin’<sup>16</sup>)
- p. λαγχάνω ‘obtain by lot, come to possess’ Hom.+ <sup>(A)</sup>, intr. ‘fall to one’s lot’ Hom.+: ἔλαχον Hom.+, ἐλόγχα Hom.
- q. λαμβάνω ‘take hold, grasp, seize’ Att. Ion. <sup>A</sup> (: λάζομαι Hom., Dial.): ἔλαβον Hom.+, εἴληφα Att./ἐλάβηκα Ion., Dial.<sup>\$</sup>

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a backformation from an innovative aorist to present οἰδάνω. Similar considerations hold for immediately following transitive and intransitive active κῦδάνω ‘be triumphant (intr., Y.42); glorify (tr., ξ73).’

<sup>15</sup>On this verb, see the recent discussion of Nikolaev (2023).

<sup>16</sup>See Fortson (2009), and further Jasanoff (2022), who proposes a special origin for this set of verbs and at least this subset of -άνω presents in Greek (and Armenian). In my view, this set is best understood as an instance of the general type under discussion, with Lat. *prehendō* as the continuation of \**g<sup>h</sup>ed<sup>b</sup>-e/o-*, Go. *duginnan* as a re-characterization of this inherited stem according to the productive Germanic type, and OIr. *ro·geinn* ‘occupies’ as the phonologically regular development of the same stem à la Fortson (2009).

It is possible to divide this material into two basic sets of forms. The first set, which includes everything but the final *άνω*-class in ex. (6), consists of verbs that are phonologically directly derivable from a thematic nasal-infix formation of the Gorbachov type made to the PIE root in question, with a subtype (a) that pairs with a root aorist or for which the aorist is not attested – viz. *φθίνω/φθίνω* Hom.+; *ἔφθιτο* Hom.+; *ἔφθιον* Hom.+; *ἔφθ(ε)ισα* Hom.+; *δύνω* Hom.+<sup>A</sup>; *ἔδυν* Hom.+ (intr.); *ἔδυσσά*, *-αμην* Hom.+; etc. – and a subtype (b) that pairs with a thematic aorist – viz. *θάλλω* Hes. *hHom.*+; *ἔθαλον* *hHom.*+, etc. The second set consists of verbs in *-άνω*, a highly productive and expansive category in both prehistoric and historical Greek. As far as can be demonstrated, the verbs found in this second set are not made to roots in a final laryngeal and so cannot at first sight be explained as the direct phonological results of the thematic nasal-infix formation in question – many if not all must be of analogical origin. Nevertheless, it is important to note that these verbs largely show the same semantics as set 1 (see further below), systematically pair with the thematic aorist (viz. *άνδάνω* Hom.+; *ἔαδον* Hom.+; etc.), as set 1 (and further as in Slavic), have a substantial number of forms that correlate with the Caland system, as set 1, and of course share a similar thematic active nasal stem formant, exactly as set 1. There is thus every reason to assume that both sets continue the same underlying type.

Across both sets of forms, these verbs regularly correlate with nasal presents in other IE languages. Some have exact or close formal and semantic matches of the type that Gorbachov outlined, while others have athematic nasal-infix cognates where they pattern semantically with the intransitive middle of the athematic nasal-infix cognate but not the transitive active, cf. (7)–(9).

(7) Close semantic and morphological equation or parallel<sup>17</sup>

- a. *φθίνω/φθίνω* ‘disappear, decrease’ Hom.+; ON *dvena* ‘disappear, decrease’, ON *dvína* ‘id.’ (: Ved. *kṣiṇáti*, *kṣiṇóti* AV, YAv. *jinā-ti*)
- b. *θάλλω* ‘blossom, flourish’ Hes. *hHom.*+; MW *deillyaw\**, Alb. *dal*, *del* ‘go out’
- c. *ἔχανον* ‘gape, yawn’ Hom.+; OCS *zinōti* ‘gape’, OHG *ginēn* ‘yawn’, ON *gína* ‘id.’
- d. *πυνθάνομαι* ‘learn, ascertain’ Hom.+; OCS *vŭzbŭnōti*, *vŭzbŭde* ‘wake up’, Lith. *atbuñda*, *atbùsti*, *atbùdo* ‘id.’ (: OIr. *ad-boind*)
- e. *ῥανδάνω* ‘take hold’ Hom.+; Lat. *prehendō*, *-ere*, OIr. *ro-geinn* ‘occupies’, Go. *duginnan* ‘begin’ (cf. Fortson 2009; Jasanoff 2022)

<sup>17</sup>A likely further instance of this type underlies *ὁ θρόμβος* ‘lump, clot’ Att. Ion.: Lith. *driṁba*, *dribti* ‘fall in flakes, drop’.

- (8) Semantic correspondence with athematic nasal-infix middle
- κᾰμνω ‘(tr.) produce/acquire through labor/toil (Hom.[+], but rare); (intr.) labor, toil, grow weary laboring/toiling’ Hom.+ : *śamnīṣe* ‘labor’ YV, *śamāyāte* ‘id.’ RV, *aśamīt* ‘ist ruhig geworden’ AVP, *ásamanta* ‘la-bored’ YV
  - ἀνδάνω ‘be/become pleasing’ Hom.+ : ?Ved. *s<sub>(u)</sub>vádate* ‘wird schmackhaft’ RV<sup>18</sup> (: Ved. *s<sub>(u)</sub>vádati* ‘macht schmackhaft’ RV)
  - οἰδάνω ‘swell (intr.)’ I646 (οἰδάνεται), Ar.Pax.116 (οἰδάνων); ‘make swell’ (tr.) I554 (≈ A.R.1.478): ClArm. *aytnowm* ‘swell [intr. *tantum*]
- (9) Semantic mismatch with athematic nasal-infix active<sup>19</sup>
- φθίνω/φθίνω ‘disappear, decrease’ Hom.+ : Ved. *kṣiṇāti*, *kṣiṇóti* ‘de-destroy’ AV, YAv. *jinā-ti* ‘id.’ (: ON *dvena* ‘disappear’, decrease’, ON *dvína* ‘id.’)
  - θύνω ‘rush, dart along’: Ved. *dhūnóti* ‘shake’
  - ἀνδάνω ‘be/become pleasing’ Hom.+ : ?Ved. *s<sub>(u)</sub>vádati* ‘macht schmackhaft’ RV, *svādate* ‘wird schmackhaft’ RV
  - ὀλισθάνω ‘slip’ Att. (: ὀλισθαίνω ‘id.’ Arist.+): ὤλισθε Hom.+ : Go. *slindan* ‘schlingen’ (: Ved. *srédhati* RV, *sridhat* RV)
  - πυνθάνομαι ‘learn, ascertain’ Hom.+ (: πεύθομαι Hom.+): OIr. *ad-boind* (: OCS *vŭzbŭnŭti*, *vŭzbŭde* ‘wake up’, Lith. *atbuñda*, *atbŭsti*, *atbŭdo* ‘id.’)

The semantic patterning here aligns closely with the fact that nearly two-thirds of the forms in Greek are intransitive and that transitive verbs are largely limited to cases where an original accusative of respect or goal is likely, such as λανθάνω, -ομαι ‘escape notice (±acc.); (mid.) forget’ (: λήθω, -ομαι Hom.+) or φθάνω/φθάνω ‘(tr.) be beforehand, outstrip, anticipate; (intr.) come/act first (±part.)’ Hom.+, or where the root denotes a cognitive-perceptual or possessional state, state-oriented semantic categories that are regularly transitive cross-linguistically – e.g., πυνθάνομαι ‘learn, ascertain’ Hom.+ (: πεύθομαι Hom.+) or χανδάνω ‘take hold of’ Hom.+.<sup>20</sup>

<sup>18</sup> Assuming this form continues a nasal-infix present.

<sup>19</sup> For the pair φθάνω/φθάνω ‘(tr.) be beforehand, outstrip, anticipate; (intr.) come/act first (±part.)’ Hom.+ : Ved. opt. *daghnyāt* ‘würde knapp verfehlen’, it is important to note that active *nu*-presents are often intransitive in Vedic and in some cases replace the thematic nasal-infix intransitive actives discussed here – viz. Ved. *tṛmpāti* ‘take delight in’ RV: *tṛmpóti* RV ‘id.’

<sup>20</sup> On state-oriented semantic subclasses, see Rau 2013, 2016, and 2017.

The traditional approach to these verbs, which stretches back to the 19<sup>th</sup> c. and is followed by all modern scholars, including Klingenschmitt 1982, Rix 1992, LIV<sup>2</sup> *passim*, and others, explains them as the result of an inner-Greek functionless thematization. This is problematic on all fronts and can be excluded.<sup>21</sup>

<sup>21</sup>The traditional approach, which explains the whole type this way and especially forms with suffixal  $-\acute{\alpha}\nu\omega \Leftarrow -C\eta h_x ont(i)$  bzw.  $-C\eta\upsilon ont(i)$ , has four fatal issues:

1. As noted in the survey, the Greek forms show a fundamental and unmotivated semantic mismatch with active athematic nasal-infix presents, which are in effect exclusively transitive in the languages and mostly make causative change of state/location ( $\pm$ activity) formations – apart from late NPIE  $*d^h\eta snéu^{-ti}$  (: Ved.  $d\eta\eta snóti$  ‘dare’, etc.) and some *nu*-presents in Indic and Iranian, which are obviously secondary. On the general semantic profile of nasal-infix presents, see Meiser 1993.
2. There is no reason to assume that Greek generalized the thematic 3<sup>rd</sup> pl. in  $*-ont(i)$  to athematic stems and positive evidence against this assumption. See Peters 2004 and further Willi 2018: 286–298.
3. There is no reason to assume that the syllabification of the 3<sup>rd</sup> pl. of the nasal-infix presents in Greek or the other IE languages was  $-C\eta h_x ont(i)$  bzw.  $-C\eta\upsilon ont(i)$ . The nasal-infix formant fails to show syllabic variants in its *inflectional* paradigm (see Praust 2004), at least in the environments in question here – viz. when infixes before a sonorant or laryngeal in a minimally tri-consonantal root – and instead developed an epenthetic vowel, a situation also evidenced by the Greek treatment of the nasal-infix presents to laryngeal final roots. The default assumption is that this also held in the 3<sup>rd</sup> pl. and that the epenthetic vowel was later deleted as in Indo-Iranian.
4. Even if these assumptions were reasonable, there is no other support for the assumption that nasal-infix presents were regularly or even occasionally subject to thematization in the deeper prehistory of the language. Nasal-infix presents instead follow other well documented developmental pathways, see Sturm 2021.
  - a) Some have replaced their athematic forms with *grbhāyāti*-derivatives – e.g.,  $\acute{\upsilon}\phi\acute{\alpha}\iota\nu\omega$  ‘weave’ (: Ved. *unābdhi* ‘id.’),  $\phi\acute{\alpha}\iota\nu\omega$  ‘show, make appear’ (: ClArm. *banam* ‘disclose, discover’),  $\iota\acute{\alpha}\iota\nu\omega$  ‘warm, cheer’ (possible athematic  $\iota\nu\acute{\alpha}\omega$  ‘empty’) (: Ved. *iṣṇāti* ‘set in vigorous motion’),  $\kappa\lambda\acute{\iota}\nu\omega$  ‘incline’ (: YAv. *-sirinao^{-ti}* ‘lean on (tr.)’),  $\kappa\rho\acute{\iota}\nu\omega$  ‘judge’ (: Lat. *cernō*, *-ere* ‘distinguish, discern’). On the type, see Jasanoff 2003, and on the Greek forms, Kölligan 2010.
  - b) Nasal-infix stems to roots in a stop were regularly remodeled as *nu*-stems – cf.  $\zeta\epsilon\acute{\upsilon}\gamma\nu\mu\iota$  ‘yoke, join’ (: Ved. *yunākti* ‘id.’),  $\acute{\omicron}\rho\acute{\epsilon}\gamma\nu\mu\iota$  ‘stretch out’ (: Ved. *ṛnākti\** ‘id.’), etc.
  - c) Nasal-infix stems to roots in  $-h_2$  develop normally and then in Attic Ionic are restructured as *nu*-verbs – cf.  $\pi\acute{\iota}\tau\nu\eta\mu\iota$  ‘spread out’ Hom.+ (:  $\pi\epsilon\tau\acute{\alpha}\nu\nu\mu\iota$  Att.), etc.
  - d) Nasal-infix stems to roots in  $-h_3$  develop normally and then in the history of Greek are restructured as *nu*-stems, with further late remodeling in some cases – cf.  $\sigma\tau\acute{\omicron}\rho\nu\mu\iota$  ‘strew’ Hom.+ (:  $\sigma\tau\omicron\rho\acute{\epsilon}\nu\nu\mu\iota$  Eust.), etc.
  - e) Nasal-infix stems to roots in  $-h_1$  are early analogically restructured as thematic stems – cf.  $\beta\acute{\alpha}\lambda\lambda\omega$ ,  $\tau\acute{\alpha}\mu\nu\omega$ , etc.

The obvious alternative to this approach is to motivate the form and function of these verbs by aligning them with the active thematic nasal-infix inchoative type that Gorbachov has outlined for Baltic, Slavic, and Germanic. This is straightforward in the case of the forms in set 1) above – that is, everything but the  $\acute{\alpha}\nu\omega$ -presents. These can all easily be explained as direct phonological and semantic continuants of this type – viz.  $\varphi\theta\acute{\iota}\nu\omega/\varphi\theta\acute{\iota}\nu\omega$  Hom.+ <  $*d^h g^{uh} in\grave{u}-o/e-$  (:  $\varphi\theta\acute{\iota}\nu\acute{\upsilon}\theta\omega$  Hom.+; Ved.  $k\check{s}i\check{n}\acute{a}ti$ ,  $k\check{s}i\check{n}\acute{o}ti$  ‘destroy’ AV, YAv.  $j\acute{i}n\bar{a}^{ti}$  ‘id.’; ON  $dvena$  ‘disappear, decrease’, ON  $dv\acute{ina}$  ‘id.’),  $\theta\acute{\alpha}\lambda\lambda\omega$  Hes.  $hHom.$ + <  $*d^h l\grave{n}h_1-o/e-$  (: Alb. 1sg.  $dal$ ,  $del$  ‘go out’ <  $*daln\bar{o}$ ,  $dalnet$  and MW  $deillyaw^*$  <  $*daln-$ ), etc.

The forms in set 2) –  $\acute{\alpha}\nu\omega$ -presents to roots in obstruent without a demonstrable laryngeal –, which largely show the same morphology, semantics, and derivational relations as set 1) and cannot be explained as the result of an inner-Greek functionless thematization (see footnote 21), can also be aligned with this type under two straightforward assumptions. First, the suffix  $-\acute{\alpha}\nu\omega$  was extracted from roots that ended in an obstruent plus laryngeal – viz.  $-Cano/e-$  <  $*-C\eta h_x e/o-$  – and then extended to all obstruent final roots. This development can be understood as an example of the “externalization” of the nasal formant, a process commonly found in all IE languages and well-attested in Greek – cf., e.g.,  $\zeta\epsilon\upsilon\gamma\nu\bar{\upsilon}\mu\iota$  (: Ved.  $yun\acute{a}kti$ ),  $\omicron\rho\acute{\epsilon}\gamma\nu\bar{\upsilon}\mu\iota$  (: Ved.  $\check{r}\acute{n}\acute{a}kti^*$ ), etc. or, with the  $grbh\bar{a}y\acute{a}ti$ -type, Gk.  $\upsilon\phi\alpha\acute{\iota}\nu\omega$  (: Ved.  $un\acute{a}bdhi$ ),  $\kappa\bar{\upsilon}\delta\alpha\acute{\iota}\nu\omega$  ‘glorify, honor’ Hom.+ ( $\equiv$   $\iota\acute{\alpha}\acute{\iota}\nu\omega$  ‘invigorate, warm’ [: Ved.  $i\check{s}\check{n}\acute{a}ti$  ‘set in vigorous motion’]), etc. – and can be seen to be exactly parallel to what has happened in Slavic and Germanic, where the synchronically normal shape of the formant has been extracted from sonorant-plus-laryngeal-final roots – viz.  $-I/Rne/o-$  <  $*-I/Rnh_x e/o-$  – and extended at the expense of all other stem shapes – e.g., OCS  $pril\acute{i}(p)n\check{o}$ ,  $-l\acute{i}(p)n\check{o}ti$  ‘cleave, cling to’, Go.  $aflifnan$  ‘remain’, ON  $lifna$  ‘id.’ (: Lith.  $li\check{m}pa$ ,  $li\check{p}ti$  ‘cling, stick to; climb’: Ved.  $limp\acute{a}ti$  AV+). Second, the extension of the formation to transitive verbs denoting cognitive-perceptual and possessional states, which would not a priori be expected from an anticausative formant made to causative change of state verbs of the sort found here, can be understood to result from either 1) the general reinterpretation and extension of the formant as an all-purpose change of state marker – e.g., Lith.  $supra\check{n}ta$ ,  $supra\check{s}ti$ ,

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The nasal-infix stems that otherwise show up as thematic in Greek are 1) late backformations from the aorist – cf., e.g.,  $\tau\acute{\alpha}\nu\upsilon\omega$  ‘stretch’ Hom.+ (:  $\tau\acute{\alpha}\nu\upsilon\mu\alpha\iota$  Hom.),  $\acute{\alpha}\nu\upsilon\omega$  ‘seek, strive’ Hom.+ (:  $\acute{\alpha}\nu\upsilon\mu\epsilon\varsigma$  Theoc.+), 2) of denominative origin – cf., e.g.,  $\delta\acute{\iota}\nu\acute{\epsilon}\omega$  ‘whirl around’,  $\kappa\acute{\iota}\nu\acute{\epsilon}\omega$  ‘set in motion’, on which see Nikolaev 2025a,b – or obviously secondary –  $\acute{\iota}\kappa\nu\acute{\epsilon}\omicron\mu\alpha\iota$  ‘reach, arrive’ Hom.+ Att. Ion. (:  $\acute{\iota}\kappa\omega$  Hom. Dial.) –, 3) likely due to reinterpretation of archaic volitional/desiderative subjunctives as indicatives in performative, speech-act contexts *et sim.* – viz.  $\omicron\mu\nu\acute{\upsilon}\omega$  ‘swear’ Hom.+ Dial. (:  $\omicron\mu\nu\bar{\upsilon}\mu\iota$  Hom.+),  $\tau\acute{\iota}\nu\omega/\tau\acute{\iota}\nu\omega$  ‘pay’ Hom.+ (: < $\tau\acute{\iota}\nu\nu\nu\alpha(\iota)$ > Kyme, ca. 700 BCE, Bartoněk & Buchner 1997: 201–214,  $\tau\acute{\iota}\nu\upsilon\mu\alpha\iota$  Hom. Hes. Hdt. E.) –, or 4) simply morphologically obscure – viz.  $\acute{\alpha}\rho\nu\acute{\epsilon}\omicron\mu\alpha\iota$  ‘deny’ Hom.+;  $\kappa\nu\nu\acute{\epsilon}\omega$  ‘kiss’ Hom.+.

*suprāto* ‘understand’ (: Go. *fraþjan* ‘id.’) – and/or 2) the precocious generalization of the suffix as a regular present-stem formant for (state-oriented) thematic aorists (and stative/presential perfects), as normal in later Greek.<sup>22</sup>

## 4 Indo-Iranian

The type is also unambiguously attested for Indic and indirectly for Iranian – though here less robustly than for Greek.<sup>23</sup> For the nasal presents under discussion, see further Kulikov 2000, Hill 2007, and especially Zasada 2021 s.vv., who provides critical discussion of the relevant forms. The forms can usefully be divided into finite and participial examples.

### (10) Intransitive nasal-infix thematic actives in Vedic: Finite Forms

- a. *tr̥mpāti* ‘enjoy, take delight in, fill up on (+gen.)’ RV (12x), *tr̥pnóti* ‘id.’ RV (5x), *tr̥pyati* AV+: *át̥rpat* AVŚ 3.13.6 (: adv. neut. *tr̥pát* RV 3x); cf. YAv. n. sg. *θraǵs̥* ‘satisfaction (*vel sim.*)’ Yt.5.26, YAv. *θraǵəδra-* ‘rich in; satisfied’, which points to the Indo-Iranian age of the present.
- b. *śrathnās*, *śrathnité* ‘loosen, make slack’ RV+: *śrathayanta* ‘loosen, become slack’.

RV 2.24.3ab:

*tād devānām devátamāya kártuvam áśrathnan*  
 that.NOM god.GEN.PL god.SUPERL.INS.SG do.GER loosen.3PL.IPF.ACT  
*d̥ṛl̥hā<sup>+</sup> ávradanta vīlitā* |  
 firm.NOM.PL.N soften.3PL.IPF.MID hard.NOM.PL.

‘That had to be done by the foremost god of gods: what was firm became loose, what was hard became pliant.’ (Jamison & Brereton 2014)

Cf. TS 6.1.9.7 *ánu śrathati* ‘become loose.’ On the geographic localization of the forms of the root with nasal, see Insler 1991: 95<sup>3</sup>.

- c. *r̥nóti*, *r̥nváti* ‘set in motion, impel (tr., intr. mid.)’, *íyarti*, *írate* ‘id.’ RV+: *ārta* (intr.) RV.+.

<sup>22</sup>From this starting point the type will then have been fully morphologized as the regular present stem correspondent to thematic aorists and then further extended. A similar process will have taken place in Armenian.

<sup>23</sup>Possible indirect evidence for the type may also survive in the RV and AV in the tendency noted by Kulikov 2000 for nasal-infix verbs that have relatively well attested athematic and thematic by-forms to show a preference for the thematic variant in intransitive usages.

RV 6.2.6ab:

*tveṣás*                      *te*                      *dhūmá*                      *r̥ṇvati*  
 turbulent.NOM.SG you.2SG.GEN smoke.NOM.SG impel.3SG.PRS.ACT  
*diví*                      *śāñ*                                              *chukrá*  
 heaven.LOC.SG be.PRS.PTCP.ACT.NOM.SG gleaming.NOM.SG  
*átataḥ* |  
 stretched.out.NOM.SG

‘Your smoke, when it is in heaven, is turbulent in motion, stretched out (there) gleaming, ...’ (Jamison & Brereton 2014).

Possibly further RV 1.144.5cd, 3.11.2c.

- d. *jínóti/te, jínvati/te* ‘revitalize, refresh (tr., intr. mid.)’ RV+.

AVŚ 12.1.46cde:

*krimir*                      *jínvat*                                              *pr̥thivi*                      *yádyad*  
 worm.NOM.SG revitalize.3SG.IPF.ACT Earth.VOC.SG whichever  
*ējati*                                              *prāvṛṣi*                                              *tán*                      *naḥ*  
 move.3SG.PRS.ACT rain.season.LOC.SG that.NOM.SG us.1PL.ACC  
*sárpan*                                                                      *mópa*                                              *sṛpat*  
 crawling.PRS.PTCP.NOM.SG.ACT NEG+to.towards.PRVB crawl.3SG.AOR.ACT

‘Welcher auflebende Wurm auch immer, o Erde, sich in der Regenzeit regt, der möge, wenn er kriecht, nicht zu uns herankriechen.’ (Zasada 2021: 105).

Possibly also AVŚ 11.4.14c, 16d = AVP 16.22.4b, 6d, AVŚ 12.1.3c. Cf. mid. 3<sup>rd</sup> sg. *jinvate* (intr.) RV 3.2.11ab. See Zasada 2021: 104ff.

- e. *śvīndate* ‘shine’ Dhātup. (: *áśvitan*, *áśvait*, *śvītāná-* RV) (: ORuss. *svī(t)nuti* ‘grow light’, Lith. *šviñta*, *švisti*, *švito* ‘dawn’, ON *hvitna* ‘goes pale’).<sup>24</sup>

# (11) Intransitive nasal-infix thematic actives in Vedic: Participial forms

- a. *siñcāti/te* ‘pour out (tr., intr. mid.)’ RV+ (: YAv. *-hiñca-ti* ‘id.’): *sécate* ‘pour out (intr.)’ RV 10.96.1 (: OAv. PN *Haēcat̪.aspa-*), *sicyáte* ‘id.’: RV *ásicat/ta* RV+.

<sup>24</sup>Given its predominantly active but semantically intransitive aorist forms, which points to a verbal root primarily lexified as an intransitive, I tentatively include this verb here, despite its likely finite middle inflection, which can easily be explained as analogical bzw. pleonastic, as outlined below.

RV 5.85.6cd:

*ékaṃ yád udná ná pṛṇánti*  
 single.ACC.M which water.INSTR.SG NEG fill.3PL.PRS.ACT  
*énīr ā-siñcántīr*  
 mottled.NOM.PL.F PRVB-pouring.PRS.PTCP.ACT.NOM.PL.F  
*avánayaḥ samudráṃ ||*  
 stream.NOM.PL.F sea.ACC.SG.M

‘That the mottled streams, pouring out, do not fill the single sea with water.’ (Jamison & Brereton 2014).

Possible also RV 1.121.6cd (*siñcáñ*). See Hill 2007: 92–95 with lit.

- b. *prá ... ṛñjánti* ‘stretch out (tr.)’, 3<sup>rd</sup> pl. mid. *ṛñjáte* ‘stretch oneself out; move quickly forwards’, them. *ṛñjate* ‘id.’: *abhí ... ṛjyate* RV 1.140.2 ‘id.’.

RV 1.95.7ab:

*úd yaṃyamīti savitéva bāhú ubhé*  
 up raise.3SG.PRS.INT.ACT Savitar.like arm.ACC.DU.M both.ACC.DU.F  
*sícau yatate bhīmá*  
 seam.ACC.DU.F fearsome.NOM.SG.M moving.towards.3SG.PRS.MID  
*ṛñján |*  
 charge.on.PRS.PTCP.ACT.NOM.SG.M

‘Like Savitar, he raises up his arms again. Healing himself along the two seams (of the world?), the fearsome one charging straight on.’ (Jamison & Brereton 2014).

RV 1.172.2abc:

*āré sá vaḥ sudānavo |*  
 distance.LOC DEM.PRON.NOM.SG.F you.GEN.DAT.PL drop.rich.VOC.PL  
*máruta ṛñjatí śáruḥ |*  
 Marut.VOC.PL moving.straight.PRS.PTCP.NOM.SG.F arrow.NOM.SG.F  
*āré ásmā yám ásyatha ||*  
 distance.LOC stone.NOM.SG which.ACC hurl.2PL.PRS.ACT

‘In the distance be your straight-aiming arrow, you Maruts rich in drops, in the distance the stone that you hurl.’ (Jamison & Brereton 2014).<sup>25</sup>

<sup>25</sup> Given its intransitive semantics, I include this form here; its athematic inflection represents a trivial inner-Vedic innovation.

RV 3.31.1:

*śāsad* *vāhnir* *duhitúr*  
 instruct.PRS.PTCP.ACT.NOM.SG.M conveyor.NOM.SG.M daughter.GEN.SG  
*naptíyaṃ* *gād*  
 (grand)daughter.ACC.SG come.3SG.AOR.INJ.ACT  
*vidvāṃ* *ṛtāsya* *dīdhitim*  
 knowing.PRF.PTCP.ACT.NOM.SG.M truth.GEN.SG contemplation.ACC.SG  
*saparyán* | *pitā* *yātra duhitúḥ*  
 serving.PRS.PTCP.ACT.NOM.SG.M father.NOM.SG where daughter.GEN.SG  
*sékam* *ṛñján* *sám*  
 outpouring.ACC.SG stretching.out.PRS.PTCP.ACT.NOM.SG together  
*śagmíyena* *mānasā* *dadhanvé* ||  
 capable.INS.SG mind.INS.SG run.3SG.PRF.MID

‘Instructing the (grand)daughter [= Āhavanīya?] of his daughter [= Gārhapatya?], the (offering-)conveyor [= Agni] has come – he the knowing one, ritually serving the visionary power of truth – to where the father [= priest? Agni?], stretching out straight, has run toward the outpouring of his daughter [=butter offering?] with capable mind.’ (Jamison & Brereton 2014).

RV 4.38.7,8:

*utá syá* *vājí* *sáhurir*  
 and DEM.PRON.NOM.SG.M prizewinner.NOM.SG.M victorious.NOM.SG.M  
*ṛtāvā* *śúśrūsamāṇas* *tanúvā*  
 truthful.NOM.SG.M fame.seeking.DES.PCTP.MID.NOM.SG.M body.INSTR  
*samaryé* | *túraṃ* *yatíṣu*  
 clash.LOC headlong.ADV go.PRS.PTCP.ACT.LOC.PL.F  
*turáyann* *ṛjipyó* *ádhi*  
 hasten.PRS.ACT.PCTP.NOM.SG.M hurrying.NOM.SG.M towards  
*bhruvóḥ* *kirate* *reṇúm*  
 eyebrows.LOC.DU.F scatter.PRS.MID.3SG dust.ACC  
*ṛñján* ||  
 stretch.out.PRS.PTCP.ACT.NOM.SG.M

*utá sma asya* *tanyatór* *iva dyór*  
 and EMPH DEM.PRON.GEN.SG.M thunder.ABL like heaven.GEN  
*ṛghāyató* *abhiyújo* *bhayante* | *yadā*  
 tremble.PRS.PCTP.ACT.GEN.SG charge.ABL fear.PRS.MID.3PL when  
*sahásram abhí ṣīm* *áyodhīd*  
 thousand to PERS.PRON.ACC.SG battle.AOR.ACT.3SG

*durvartuḥ* *smā bhavati*  
 difficult.to.obstruct.NOM.SG.M EMPH become.PRS.ACT.3SG  
*bhīmá* *rñján* ||  
 fearsome.NOM.SG.M stretch.out.PRS.PTCP.ACT.NOM.SG.M

‘And this prizewinner, victorious, truthful, himself seeking fame with his own body in the clash, hastening headlong toward them (fem. [=mares?]) as they go hastily, straight-flying, scatters dust up to the eyebrows as he stretches out straight. And they take fear at his charge as he shows his mettle, as if at the thundering of heaven. When a thousand have battled him, the fearsome one becomes difficult to obstruct, as he stretches out straight.’ (Jamison & Brereton 2014).

- c. *ṛṇádhat* ‘accomplish’ RV AV, *ṛdhnóti* ‘id.’ RV 1x, AV 1x:  
*ṛdhyate/ṛdhyáte* ‘succeed’ RV+.

RV 1.173.11:

*yajñó* *hí śma índaram<sup>+</sup> kás cid*  
 sacrifice.NOM.SG.M for EMPH Indra.ACC whichever.NOM.SG.M  
*ṛndhāñ* *juhurāñás*  
 succeeding.PRS.PTCP.ACT.NOM.SG.M swerving.AOR.PTCP.MID.NOM.SG.M  
*cin mánasā pariýán* | *tīrthé ná*  
 EMPH mind.INS.SG go.around.PRS.PTCP.ACT.NOM.SG.M ford.LOC.SG as  
*áchā tāṭṛṣāṇám óko dīrghó ná*  
 to thirst.PRF.PTCP.MID.ACC.SG.M home.ACC.SG.N long.NOM.SG.M as  
*sidhrám á kṛṇoti ádhvā* ||  
 goal.ACC.SG PRVB do.3SG.PRS.ACT road.NOM.SG.M

‘For any sacrifice that reaches fulfillment, even though it swerves along, meandering in mind, brings Indra to the house, as if bringing a thirsting man to a ford – as a long road brings home a man who reaches his goal.’ (Jamison & Brereton 2014).

- d. *mināti/te* ‘damage, harm; exchange, change (tr., intr. mid.)’ RV+ (: *míyate* ‘disappear, pass away’ RV).

RV 5.2.1cd:

*ánikam asya ná mináj*  
 face.NOM.SG.N his NEG changing.PRS.PTCP.ACT.NOM.SG.N  
*jānāsaḥ puráh paśyanti níhitam arataú* |  
 people.NOM.PL front see.3PL.PRS.ACT down.set.ACC.SG.N spokes.LOC

‘His face is not one that changes (its face): the peoples see it in front, set down in the circle of spokes.’ (Jamison & Brereton 2014).

Cf. intr. them. mid.:

RV 1.79.2a:

*á*      *te*                      *suparṇá*                      *aminantaṃ*  
PRVB your.GEN.SG fine.feathered.NOM.PL.M change.3PL.IPF.MID  
*évaiḥ*  
way.INS.PL

‘Your fine-feathered (lightning flashes) zigzagged along their ways.’  
(Jamison & Brereton 2014).

- e. *muñcāti/te* ‘release, set free (tr., intr. mid.)’ RV+: *múcyate/mucyáte* ‘get free’ RV+ (: *ámucat* RV+, *ámugdhvam* RV+; OCS *promŭknŭ*, *-mŭknŭti sę*, Lith. *muñka*, *mŭkti*).<sup>26</sup>

AVŚ 8.7.10ab:

*unmuñcántīr*                      *vivaruṇá*                      *ugrá*  
PRVB.get.free.PRS.PTCP.NOM.PL.F off.Varuṇa.NOM.PL.F mighty.NOM.PL.F  
*yá*                      *viṣadúṣanīḥ* |  
who.NOM.PL.F destroying.poison.NOM.PL.F

‘Sich befreiend, frei von Varuṇa, mächtig, die Gift zerstörend sind... [= Pflanzen].’ (Zasada 2021: 214).

Cf. 8.7.4 (intr.) *prastrṇatīś*, *pratanvatīś*. For full discussion, see Zasada 2021: 214.

- f. *ásinvant-* ‘insatiable’ RV (5x; *asinvá-* ‘id.’ RV) if as likely to tr. *\*sinóti* ‘satisfy’ (: TA *siṃseñc* ‘satisfy’, mid. *siṃsantār* ‘become satisfied’, TB *sinastar* ‘id.’).

RV 10.79.1d:

*ásinvatī*                      *bápsatī*                      *bhūri*  
insatiable.NOM.DU.F gnawing.PRS.PTCP.NOM.DU.F much.ACC.SG  
*attaḥ*  
eat.3DU.PRS.ACT

‘Insatiable, gnawing, they eat amply.’ (Jamison & Brereton 2014).

There are two interesting observations that can be made about this material.<sup>27</sup> First, there is a fundamental difference in the distributional attestation of this

<sup>26</sup>Note further OP <a-m<sup>u</sup>-u-θ> /amunθa/ D I 01, ii 2, 71, iii 71 ‘fled, escaped’, which as generously pointed out by a reviewer is most easily taken as an intransitive thematic imperfect.

<sup>27</sup>It is also important to note that this type regularly correlates with the thematic aorist in Indic and Iranian – cf., e.g., *tṛmpāti* ‘enjoy, take delight in, fill up on (+gen.)’ RV: *átrpat* AVŚ 3.13.6 (: adv. neut. *trpát* RV) or *siñcāti/te* ‘pour out (tr., intr. mid.)’ RV+ (: YAv. *-hiñca-ti* ‘id.’); RV *ásicat/ta* RV+ – as regularly in Greek (and Armenian) and systematically in OCS. This is obviously an inherited feature of a subset of the verbs in this class.

type in Greek and Indo-Iranian. In Greek, nearly all the forms we find are independent lexical items that synchronically lack the corresponding transitive nasal-infix form from which they have been derived. In Indo-Iranian, while there are such cases, we mostly find that the forms are independent, one-off active thematic intransitive uses of an otherwise well-attested nasal stem that normally signals causativity alternations with active vs. middle voice. This distribution suggests that the derivational type survived as a living process into Indo-Iranian and then early Indic at least in some verbs and/or as a dialectal phenomenon. What we have in our texts should thus likely be considered a mix of archaic and/or dialect forms and poetic innovations based on them.<sup>28</sup>

Table 4: Derivational & chronological relationship between athematic and thematic nasal-infix stems

| Nasal-Infix Causative                                                                                                            |   | Nasal-Infix Anticausative                                                               |
|----------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------|
| PIIr. <sub>1</sub> * <i>sinákti</i> , <i>sinktái</i> ‘pour out (tr., intr.)’                                                     | → | * <i>sinčáti</i> ‘pour out (intr.)’                                                     |
|                                                                                                                                  |   | ↓                                                                                       |
| ↓                                                                                                                                |   | * <i>sinčáti</i> , <i>sinčátai</i> ‘id.’                                                |
|                                                                                                                                  |   | ↓                                                                                       |
| PIIr. <sub>3</sub> * <i>sinčáti</i> , <i>sinčátai</i> ‘pour out (tr., intr.)’ (> Ved. <i>siñcáti/te</i> , YAv. <i>hiñca-ti</i> ) | ← | * <i>sinčátai</i> ‘id.’ (residual<br>* <i>sinčánt-</i> > Ved. <i>siñcánt-</i><br>‘id.’) |

The second thing to note is that the preservation of this type into Indo-Iranian provides a straightforward explanation of another remarkable feature of some of the verbs that show this usage – the fact that already in the RV and in some cases in Avestan they are fully thematized, without any trace of their original athematic inflection – viz. *siñcáti/te* RV+: YAv. *-hiñca-ti*, *muñcáti/te* RV+ (: OCS *promŭknŏ*, *-mŭknŏti se*, Lith. *muñka*, *mùkti*). For the attestation of these forms and others like it, see Hill 2007. The highly precocious thematization we find here and in similar cases – so, e.g., *kṛntáti/te* RV+, YAv. *kərənta-ti/te* (: OHG *scrintan*) and other stems with obstruent final roots liable to the causative alteration – makes perfect sense under the assumption that in some cases the thematic intransitive active was subject to an early pleonastic medialization, which then led to the backformation of a thematic transitive active and the displacement of the original athematic stem. The full derivational and chronological relationship between these forms is given in Table 4.

<sup>28</sup>The poetic character of the type can be seen clearly in the formulaic-template-type usage of, e.g., *pāda*-final *ṛñján*.

Both analogical developments are well paralleled in Indic and Iranian – cf., e.g., Ved. *édhate* ‘flourish, increase’ (: αἶθω ‘burn (intr.)’ Pi.S., αἶθουσα ‘portico’ Hom.+; αἶθομαι Hom.+, ClArm. *ayrem* ‘burn’), for “pleonastic” medialization, and Ved. *śráyati* ‘lean on (tr.)’ (: YAv. *-sirinao-<sup>ti</sup>*; Ved. *śráyate*), for the backformation of a transitive active from an intransitive middle stem.<sup>29</sup>

## 5 Conclusion

Indo-Iranian and Ancient Greek<sup>30</sup> thus provide a very close parallel to the type outlined by Gorbachov for the Northern Indo-European languages – one that matches it morphologically, semantically, and derivationally. Based on the close agreement of these languages, it is possible to confirm Gorbachov’s analysis of the Northern Indo-European material and to reconstruct the formation for late Nuclear Proto-Indo-European as an important deverbal and deadjectival intransitive change of state type.<sup>31</sup>

## Abbreviations

|           |                    |          |               |
|-----------|--------------------|----------|---------------|
| A.R.      | Apollonius Rhodius | E.       | Euripides     |
| Alb.      | Albanian           | Eust.    | Eustathius    |
| Ar.       | Aristophanes       | Gal.     | Galen         |
| Arc.      | Arcadian           | Gk.      | Greek         |
| Archil.   | Archilochus        | Go.      | Gothic        |
| Arist.    | Aristotle          | Hdt.     | Herodotus     |
| Att. Ion. | Attic Ionic        | Hes.     | Hesiod        |
| AV        | Atharvaveda        | Hes. Sc. | Hesiod Scutum |
| AVP       | Atharvaveda        | hHom.    | Homeric Hymn  |
|           | Paippalada         | Hitt.    | Hittite       |
| AVS       | Atharvaveda        | Hom.     | Homer         |
|           | Saunakiya          | Hp.      | Hippocrates   |
| Call.     | Callimachus        | Hsch.    | Hesychius     |
| ClArm.    | Classical Armenian | Intr.    | Intransitive  |
| Dhātup.   | Dhātupātha         | KS       | Katha Samhita |
| Dial.     | Dialects           | Lat.     | Latin         |

<sup>29</sup>See further footnote 14.

<sup>30</sup>Further Classical Armenian. See footnote 5.

<sup>31</sup>The type is also attested for Italo-Celtic as Gorbachov (2007: 217), following others, rightly notes – cf., e.g., Lat. *\*cumbō*, *-ere* Pl.+, *ning(u)it* Lucr.+, and possibly *\*hendō*, *-ere* Pl.+. A task for the future is to identify further instances in these branches.

|        |                     |         |                    |
|--------|---------------------|---------|--------------------|
| Lesb.  | Lesbian             | Pi.     | Pindar             |
| Lith.  | Lithuanian          | Pl.     | Plautus            |
| Lucr.  | Lucretius           | PN      | Proper name        |
| MW     | Middle Welsh        | RV      | Rigveda            |
| Myc.   | Mycenaean           | S.      | Sophocles          |
| OAv.   | Old Avestan         | Sa.     | Sappho             |
| OCS    | Old Church Slavonic | TA      | Tocharian A        |
| OE     | Old English         | TB      | Tocharian B        |
| OHG    | Old High German     | Th.     | Thucydides         |
| OIr.   | Old Irish           | Theocr. | Theocritus         |
| ON     | Old Norse           | Tr.     | Transitive         |
| OP     | Old Persian         | TS      | Taittiriya Samhita |
| OPr.   | Old Prussian        | Ved.    | Vedic Sanskrit     |
| ORuss. | Old Russian         | YAv.    | Young Avestan      |
| PGmc.  | Proto-Germanic      | YV      | Yajurveda          |

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